Q Learning

Thursday, June 26, 2025 3:04 PM

We assume that there is a behavioral policy π under which we observe some data $D = \{(s_0, a_0, r_0), (s_1, a_1, r_1), \dots, (s_{n-1}, a_{n-1}, r_{n-1})\}$,

where $a_i \sim \pi(\cdot \mid s_i)$, $r_i = r(s_i, a_i)$. From this data, we want to estimate the Q-function.

Recall that Q satisfies the equation

$$Q = T_Q Q,$$

where

$$(T_Q Q)_{s,a} = r(s,a) + \alpha\, E\left(\max_{a'} Q(s_1, a') \,\Big|\, \begin{matrix} s_0 = s, \\ a_0 = a \end{matrix}\right)$$

where the expectation is w.r.t. the probability transition matrix of the MDP, i.e.

it is equal to $\sum_{s'} P(s' \mid s, a) \max_a Q(s', a)$

Motivated by stochastic approximation, consider the following estimate for Q:

$$Q_{k+1}(s,a) = \begin{cases} Q_{k+1}(s,a) & \text{if } \underbrace{(s_k, a_k)}_{\text{from the dataset } D} \neq (s,a) \\ (1-\varepsilon_k) Q_k(s,a) + \varepsilon_k\left(r(s,a) + \alpha \max_{a'} Q_k(s_{k+1}, a)\right) & \text{else} \end{cases}$$

More compactly,

$$Q_{k+1}(s,a) = Q_k(s,a) - \varepsilon_k I_{(s,a)=(s_k,a_k)} \left(Q_k(s,a) - r(s,a) - \alpha \max_{a'} Q_k(s_{k+1}, a)\right)$$

Now, we make a modeling choice to analyze the convergence of the algorithm: instead of using the k^{th} element, we randomly pick a (state, action, reward, next state) from the dataset. This is somewhat similar to what is done in practice (called experience replay, but this is a simplified version). This means that if the number of elements in the dataset (n) is large, then by the Markov law of large numbers, approximately (s_k, a_k, s_{k+1}) are chosen independently at random according to

$$s_k \sim \tilde{d}_\pi \text{ (the stationary distribution of states under the policy } \pi),$$

$$a_k \sim \pi(\cdot \mid s_k)$$

and $s_{k+1} \sim P(\cdot \mid s_k, a_k)$.

Of course, the validity of the approximation depends on whether n is large enough, which depends on the Markov chain generated by the policy π. But we won't consider these details in the analysis here.

For later convenience, we define

$d_\pi(s,a) = \tilde{d}_\pi(s)\, \pi(a \mid s)$, i.e., the stationary distribution over states and actions

Lemma: $Q_k(s,a)$ is bounded $\forall\, k, s, a$ in terms of Q_0. $\square$

We will prove this lemma later, but we will assume for now that

$$|Q_k(s,a)| \le M \quad \forall\, k, s, a$$

Let $F_{k-1} = \{(s_0, a_0, r_0), \dots, (s_{k-1}, a_{k-1}, r_{k-1}), s'\}$, where $s' \sim P(\cdot \mid s_{k-1}, a_{k-1})$,

i.e, F_k is the past data, Then

$$E\left(I_{(s,a)=(s_k,a_k)}\left(Q_k(s,a) - r(s,a) - \alpha \max_{a'} Q_k(s_{k+1}, a')\right) \Big| F_k\right)$$

$$= d_\pi(s,a)\left(Q_k(s,a) - r(s,a) - \alpha \sum_{s'} \max_{a'} Q_k(s', a')\, P(s' \mid s, a)\right)$$

$$= d_\pi(s,a)\, (T_Q Q)(s,a)$$

In vector form,

$$Q_{k+1} = Q_k - \varepsilon_k D_\pi (Q_k - T_Q Q_k - W_k)$$

where $D_\pi = \operatorname{diag}(d_\pi)$ and W_k is the difference between the true value and the expectation since Q is bounded, $\{W_k\}$ is bounded,

$E(W_k \mid F_{k-1}) = 0$

$$\Rightarrow E\left(\|D_\pi W_k\|_p^2\right) \le \sigma^2 \quad \forall k$$

Recall that SA for contraction mappings was of the form:

$$x_{k+1} = x_k + \varepsilon_k (T(x_k) - x_k - w_k)$$

$$\Rightarrow Q_{k+1} = Q_k + \varepsilon_k (Q_k - D_\pi Q_k - Q_k + D_\pi T_Q Q_k + D_\pi W_k)$$

Or $Q_{k+1} = Q_k + \varepsilon_k (Q_k - H(Q_k) + D_\pi W_k)$

So to apply that result, we need to show that

$D_\pi Q - D_\pi T_Q Q + Q =: H(Q)$ is a contraction

$$H(Q) = (D_\pi - I) Q + D_\pi T_Q(Q)$$

$$H(Q_1) - H(Q_2) = (D_\pi - I)(Q_1 - Q_2) + D_\pi (T_Q(Q_1) - T_Q(Q_2))$$

$$\Rightarrow \|H(Q_1) - H(Q_2)\|_\infty$$

$$= \max_{s,a} \left| (d_\pi(s,a) - 1)(Q_1 - Q_2)(s,a) + d_\pi(s,a)(T_Q Q_1 - T_Q Q_2)(s,a) \right|$$

$$\le \max_{s,a} (1 - d_\pi(s,a)) |(Q_1 - Q_2)(s,a)| + d_\pi(s,a) |(T_Q Q_1 - T_Q Q_2)(s,a)|$$

$$\le \max_{s,a} \left((1 - d_\pi(s,a)) + \alpha\, d_\pi(s,a)\right) |Q_1(s,a) - Q_2(s,a)|$$

$$\le \underbrace{\max_{s,a} \left(1 - (1-\alpha) d_\pi(s,a)\right)}_{<1 \text{ if } \underbrace{d_\pi(s,a) > 0 \ \forall s,a}_{\Rightarrow \pi \text{ has to be sufficiently exploratory}}} \|Q_1 - Q_2\|_\infty$$

Proof of Lemma:

Let $M_0 = \max\left\{\|Q_0\|_\infty, \frac{r_{max}}{1-\gamma}\right\}$

and define $M_t = \|Q_t\|_\infty$. We will show that

$$M_t \le M_0 \quad \forall t$$

if $(s,a) = (s_t, a_t)$

$$Q_{t+1}(s,a) = (1-\varepsilon_t) Q_t(s,a) + \varepsilon_t \left(r(s,a) + \alpha \max_{a'} Q_t(s,a')\right)$$

$$\Rightarrow |Q_{t+1}(s,a)| \le (1-\varepsilon_t)|Q_t(s,a)| + \varepsilon_t (r_{max} + \alpha \|Q_t\|_\infty)$$

$$\le (1-\varepsilon_t)\|Q_t\|_\infty + \varepsilon_t (r_{max} + \alpha M_t)$$

$$= (1 - \varepsilon_t(1-\alpha))\|Q_t\|_\infty + \varepsilon_t r_{max}$$

if $(s,a) \neq (s_t, a_t)$

$$|Q_{t+1}(s,a)| = |Q_t(s,a)|$$

$$\overset{Ⓐ}{\Rightarrow} \boxed{\|Q_{t+1}\|_\infty \le \max\left\{(1-\varepsilon_t(1-\alpha))\|Q_t\|_\infty + \varepsilon_t r_{max}, \|Q_t\|_\infty\right\}}$$

$$\Rightarrow \|Q_1\|_\infty \le \max\left\{(1-\varepsilon_0(1-\alpha))\|Q_0\|_\infty + \varepsilon_0 r_{max}, \|Q_0\|_\infty\right\}$$

Recall $M_0 = \max\left\{\frac{r_{max}}{1-\alpha}, \|Q_0\|_\infty\right\}$

$$\Rightarrow M_0 \ge \frac{r_{max}}{1-\alpha} \text{ and } M_0 \ge \|Q_0\|_\infty$$

$$\Rightarrow \|Q_1\|_\infty \le \max\left\{(1-\varepsilon_0\cancel{(1-\alpha)})M_0 + \varepsilon_0 M_0\cancel{(1-\alpha)}, M_0\right\}$$

$$= M_0$$

Assume $M_t \le M_0$ for $t \ge 1$.

$$\Rightarrow M_{t+1} \underset{\substack{\uparrow \\ \text{from Ⓐ}}}{\le} \max\left\{(1-\varepsilon_t(1-\alpha))M_0 + \varepsilon_t r_{max}, M_0\right\}$$

$$\le M_0$$

$\square$